

MATH • THE FRAME IT NEVER NEEDED

Eight variants, one frame

Every slide after this one is a math slide on the shared Form frame. The running header, the footer and page number, and the **Q3 REVIEW** mark in the masthead bay are what a chrome-exempt section could never render — `meta:` did nothing at all on a math slide until this change.

The hero equation and its legend, side by side.

$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y$$

WHERE

- $\hat{\beta}$ — OLS coefficient vector
- X — design matrix, $n \times p$
- y — response vector, length n
- $X^{\top} X$ — Gram matrix, $p \times p$, must be invertible

The proof chain no longer pins its own heading.

$f(x + h) - f(x) = f'(x) h + O(h^2)$	<i>subtract $f(x)$</i>
$\frac{f(x+h)-f(x)}{h} = f'(x) + O(h)$	<i>divide by $h \neq 0$</i>
$\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} = f'(x)$	take the limit

Cards center in the stage, not under an absolute title.

DEFINITION. A function $f : [a, b] \rightarrow \mathbb{R}$ is *continuous* on $[a, b]$ if $\lim_{x \rightarrow c} f(x) = f(c)$ for every $c \in [a, b]$.

THEOREM. Let f be continuous on $[a, b]$ and let y lie strictly between $f(a)$ and $f(b)$. Then there exists $c \in (a, b)$ with $f(c) = y$.

PROOF. Set $S = \{x \in [a, b] : f(x) < y\}$. S is non-empty and bounded; let $c = \sup S$. Continuity at c forces $f(c) = y$.

The columns are the stage now, so nothing spans them.

ORDINARY LEAST SQUARES

$$\hat{\beta}_{\text{OLS}} = (X^{\top} X)^{-1} X^{\top} y$$

Unbiased under exogeneity; minimum variance among linear unbiased estimators.

RIDGE

$$\hat{\beta}_{\lambda} = (X^{\top} X + \lambda I)^{-1} X^{\top} y$$

Biased, lower variance. The penalty λ buys stability when $X^{\top} X$ is ill-conditioned.

A wrapped property no longer returns to the left margin.

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}$$

- `SHAPE` — $n \times (p + 1)$
- `ROWS` — observations
- `COLS` — intercept + p features
- `RANK` — full rank is what gives OLS a unique solution
- `COLUMN 0` — all-ones, absorbs the intercept

A label that is a symbol is set as a symbol.

$$\begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

- A — the original matrix being factorized
- L — lower-triangular, unit diagonal
- U — upper-triangular
- USE — solve $Ax = b$ by forward then back substitution

The title lands where every other title in the deck lands.

$$\hat{\beta} = 0.42 \pm 0.03$$

95% CI: [0.36, 0.48]

$p < 0.001$ · $n = 1,204$

For every additional unit of exposure the outcome rises by 0.42 SD — roughly an **8%** shift on the baseline. The effect size is the headline; the p -value only rules out chance.

The plot is sized against its cell, not against the slide.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Maps $\mathbb{R} \rightarrow (0, 1)$. *S*-shaped, $\sigma(0) = 0.5$,
steepest slope at the origin.

